



Laxatives and Anti-diarrhoeal Agents

Pharmacology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which laxative should be avoided in pregnancy because it can stimulate uterine contractions?
2. Why are bulk laxatives risky in immobile patients?
3. Which two antidiarrhoeal drugs are meperidine analogues, and what group of patients should avoid them?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Constipation & general laxative principles

Constipation: reduced frequency or ease of stool passage from a person's own normal pattern. **Non-drug therapy is first-line in all cases:** a fibre-rich diet (fruits, vegetables, wholemeal bread), increased fluid intake, minimising tea/coffee, and physical exercise to activate abdominal muscles and peristalsis.

Laxatives accelerate GI transit. That has consequences: they can **reduce the effect of poorly-absorbed, delayed-acting, or extended-release oral drugs** by speeding them through the gut before absorption completes. Chronic use risks **electrolyte imbalance** and carries a **dependency risk**.

Indications	Short-term constipation (avoid prolonged use — dependence risk); increasing GIT excretion of toxic substances; bowel prep before X-ray/colonoscopy; painful anal conditions (fissure, piles); postoperative — e.g. after haemorrhoid surgery, to avoid straining.
Contraindications	Undiagnosed abdominal pain or IBD — risk of intestinal perforation. Organic GIT obstruction.

2 Laxative classes, by mechanism

A. Irritants & stimulants	Senna: widely used stimulant laxative; oral, evacuates the bowel in 8–10h; also causes water/electrolyte secretion into the bowel. Bisacodyl: suppository or enteric-coated tablet; potent colon stimulant, acts directly on nerve fibres in the colonic mucosa. Castor oil: broken down in the small intestine to ricinoleic acid — an irritant that promptly increases peristalsis. Avoid in pregnancy — may stimulate uterine contractions.
B. Bulk laxatives	Hydrophilic colloids from indigestible parts of fruit/vegetables (methylcellulose, psyllium seeds, bran). Form gels in the large intestine → water retention and intestinal distension → increased peristaltic activity. Use cautiously in immobile patients — risk of intestinal obstruction.
C. Saline & osmotic laxatives	Magnesium citrate/hydroxide: nonabsorbable salts that hold water in the gut by osmosis → bowel distension → defecation within a few hours. Polyethylene glycol (PEG) electrolyte solutions: used as colonic lavage before radiologic/endoscopic procedures. Lactulose: semisynthetic disaccharide, not hydrolysed by GI enzymes but degraded by colonic bacteria into lactic, formic, and acetic acids → raised osmotic pressure → fluid accumulation, colon distension, soft stool, defecation.
D. Stool softeners (emollients / surfactants)	Docusate sodium/calcium — surface-active agents that emulsify with stool for softer, easier passage. Take days to work — used for prophylaxis , not acute treatment.
E. Lubricant laxatives	Mineral oil and glycerin suppositories ease passage of hard stool. Mineral oil must be taken orally in an upright position to avoid aspiration and the resulting risk of lipid pneumonia.
F. Chloride channel activators	Lubiprostone — currently the only drug in this class. Activates intestinal chloride channels → increases fluid secretion into the lumen → eases stool passage with little change in electrolyte balance. Used for chronic constipation specifically because it is not associated with tolerance or dependency , and drug-drug interactions are minimal (rapid metabolism in the stomach and jejunum).

Mnemonic — six mechanisms, A to F: “*Irritating Bulk Salts Soften Lubricated Channels.*” (Irritants/stimulants · Bulk · Saline/osmotic · Stool softeners · Lubricants · Chloride channel activators)

3 Anti-diarrhoeal agents

Diarrhoea results mainly from **increased GI motility** and **decreased fluid absorption**. Three approaches to severe acute diarrhoea: (1) maintain fluid/electrolyte balance, (2) anti-infective agents where indicated, (3) spasmolytic or other antidiarrhoeal agents.

A. Antimotility agents (diphenoxylate + atropine, loperamide)	Both are meperidine analogues with opioid-like gut action. Activate presynaptic opioid receptors in the enteric nervous system → inhibit acetylcholine release → decreased peristalsis. Lack analgesic effect at usual doses. Avoid in young children and in severe colitis — risk of toxic megacolon.
B. Adsorbents (aluminium hydroxide, activated charcoal, methylcellulose)	Presumed to work by adsorbing intestinal toxins/microorganisms and/or coating and protecting the mucosa. Much less effective than antimotility agents, and can interfere with absorption of other drugs.
C. Agents modifying fluid & electrolyte transport (bismuth subsalicylate)	Used for traveller's diarrhoea — decreases fluid secretion in the bowel, via its salicylate component and a coating action. ADR: black tongue, black stools.

P Must-remember pearls

- Non-drug measures (fibre, fluids, exercise) are always first-line for constipation — laxatives are the second step.
- Never give a laxative for undiagnosed abdominal pain, IBD, or GIT obstruction — perforation risk.
- Bulk laxatives need adequate mobility and fluids; in immobile patients they can obstruct instead of relieve.
- Diphenoxylate and loperamide are contraindicated in severe colitis and young children — toxic megacolon risk.
- Lubiprostone stands out as the one laxative not linked to tolerance or dependency.

Q Test yourself

1. A pregnant patient asks for a laxative for occasional constipation. Which one should specifically be avoided?

- A) Bisacodyl
- B) Castor oil
- C) Lactulose
- D) Docusate sodium

2. An immobile elderly patient is started on psyllium and develops worsening abdominal distension. What's the mechanism of concern?

- A) Psyllium causes disulfiram-like reactions
- B) Bulk laxatives can cause intestinal obstruction in immobile patients
- C) Psyllium is an irritant laxative
- D) Psyllium is contraindicated in the elderly regardless of mobility

3. A patient with severe active ulcerative colitis is given loperamide for diarrhoea. What is the major risk?

- A) Disulfiram-like reaction
- B) Toxic megacolon
- C) Aspiration pneumonia
- D) Black stools

4. A patient takes mineral oil lying down before bed and later develops cough and fever. What's the likely cause?

- A) Aspiration leading to lipid pneumonia
- B) An allergic reaction
- C) Electrolyte imbalance
- D) Toxic megacolon

5. A patient with chronic constipation is switched to lubiprostone and asks about the risk of becoming dependent on it. What should they be told?

- A) It carries the same dependency risk as senna
B) Lubiprostone has not been associated with tolerance or dependency
C) It cannot be used long-term for any reason
D) It requires weekly dose escalation

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1** — B — castor oil is broken down to ricinoleic acid, which can stimulate uterine contractions.
2 — B — bulk laxatives distend the bowel; in an immobile patient this raises the risk of intestinal obstruction.
3 — B — antimotility agents like loperamide are avoided in severe colitis because they can precipitate toxic megacolon.
4 — A — mineral oil taken lying down risks aspiration, which can cause lipid pneumonia; it should be taken upright.
5 — B — lubiprostone is specifically noted as not being associated with tolerance or dependency, unlike stimulant laxatives.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Susan Abdulkadir Farhadi lecture slides, LG 3.
Study alongside the lecture, not instead of it.